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Effects of Self-Questioning Training on College Students' Comprehension of Lectures

ALISON KING
Chapman College

Previous research has shown that self-questioning is an effective strategy for improving reading comprehension. The present study extended this line of research by investigating the use of self-questioning strategies with orally presented material. The study examined the relative effectiveness of four learning strategies on college students' comprehension of lectures and on their feelings of control over their success in learning. The four strategy conditions were self- and peer-questioning in small cooperative groups, independent self-questioning, review in small cooperative groups, and independent review. Students in the self-questioning conditions were trained to use a self-questioning procedure to process information presented in lecture. Results indicated that both of the self-questioning strategies significantly improved lecture comprehension over time. Significant differences were also found among the strategies with the self/peer questioners as well as the self-questioners showing post-treatment comprehension superior to that of participants using either of the review strategies. No significant change was found for students' feelings of control. These results suggest that practice in this self-questioning information-processing procedure can effectively improve college students' comprehension of lectures. © 1989 Academic Press, Inc.

For some time now there has been considerable interest in the use of self-questioning as a strategy for helping learners to understand and remember written text material. In a recent review of this topic Wong (1985) concluded that self-questioning is an effective strategy for improving reading comprehension at all levels of schooling. Students trained in generating comprehension questions during or after reading have consistently demonstrated gains in reading comprehension superior to those of students using alternative strategies such as summarizing, rereading, and answering instructor-posed questions (e.g., Davey & McBride, 1986; Nolte & Singer, 1985; Singer & Donlan, 1982; Wong, 1986).

According to one point of view, the comprehension gains resulting from self-questioning are primarily a result of the metacognitive aspects of that process (Brown, 1980; Haller, Child, & Walberg, 1988). Metacognition

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refers to an awareness of one's own cognitive processes and the self-regulation and orchestration of those processes in relation to the learning task; this includes conscious selection of strategies, matching strategy to task demands, and comprehension-monitoring (Flavell, 1981). It appears that self-questioning serves as a form of self-testing that helps the learner to monitor understanding of the material presented. Learners who use self-questioning focus on the important aspects of the material they read; they analyze the content, relate it to prior knowledge, and evaluate it in a continuous questioning–answering–questioning cycle (Nolte & Singer, 1985; Palincsar & Brown, 1984). Through this comprehension-monitoring process learners become aware of what they don't yet understand. When such learners cannot answer a question that they themselves have posed, they realize that they have failed to comprehend, and they take remedial action by asking themselves related questions, asking questions of peers, doing outside reading, or consulting other information sources (Brown & Palincsar, 1988; Markman & Gorin, 1981). From a metacognitive perspective it can be said that self-questioners know what they know and, as importantly, they know what they don't know.

Although self-questioning has been used successfully to improve reading comprehension, as yet no studies have examined the effects of a self-questioning strategy on comprehension of lectures and other oral presentations. Written text and oral communication are quite different forms of presentation; however, Danks and End (1987) have argued that reading and listening impose similar cognitive demands on the learner; and, in fact, Townsend, Carrithers, and Bever (1987) have reported that learners' reading and listening comprehension appear to be highly related. Training in self-questioning could have beneficial effects on how students process a lecture; for example, it could facilitate their organization of the lecture content, help them to relate lecture material to prior knowledge, and promote their monitoring of their understanding of the lecture content.

There is reason to expect, however, that self-questioning may not be as effective a strategy with oral material as it is with written text. For example, the process of question-asking, particularly if it occurs during a lecture, may distract the learner from attending to the lecture content. Loss of attention is of particular concern in oral contexts since the learner cannot go back and reread (or relisten) to clarify meaning or answer a question, as is possible during reading. Effective listeners use working memory capacity to manipulate information and rehearse it for storage in long-term memory. The act of generating questions, when it occurs during a lecture (and perhaps also when it occurs immediately after the lecture), may actually strain the capacity of working memory and interfere with the learner's rehearsal process, and consequently may decrease the amount of material accurately encoded into long-term memory. Therefore, self-

questioning, when used with oral material, may make greater demands on both attentional processes and on working memory than it does when used in a reading context. These additional cognitive burdens may decrease the benefits of the self-questioning strategy when used in an oral context.

The ability to comprehend material presented in lectures is critical to success in academic settings. Learning from lecture requires students to understand the information presented, store it in memory, and be able to recall it at a later time. High school and adult learners who receive so much information in lecture form would benefit from research which identifies effective strategies for learning from lecture.

Cooperative learning

If self-questioning is an effective learning strategy for individuals to use, it may be even more effective when used in a group context where learners can use each other as resources for questioning and responding about the material to be learned. Extensive research on cooperative learning in general has revealed that, at all levels of schooling and ability, subjects working in small learning groups generally outperform independent learners (Hertz-Lazarowitz, 1985; Johnson, Maruyama, Johnson, Nelson, & Skon, 1981; Sharan, 1980). One explanation for these results lies in the notion of pooled expertise. In most cooperative learning groups each learner participates to the extent possible so that collaboratively the group reaches a level of understanding and accomplishment not possible by any individual group member. The few studies which have examined effects of reciprocal questioning strategies suggest that a cooperative context may increase the effects of self-questioning. For example, Frase and Schwartz (1975) found that high school and college students who used reciprocal questioning as a study strategy scored higher on achievement tests than did those who independently reread the text. However, it is not clear how much of this effect was due to questioning per se or to the reciprocal aspects of the strategy. Fantuzzo, Dimeff, and Fox (1987) reported that college students using a reciprocal peer-questioning study strategy, in which they tested each other on course content using student-generated multiple-choice questions, received final exam scores higher than those of independent learners who simply wrote and answered questions. Fantuzzo *et al.* (1987) attributed the success of this peer-questioning strategy, at least in part, to the mutual support system inherent in the reciprocal strategy. They argued that peer equity and mutual exchange were factors which operated to maintain the feelings of support. Palincsar and Brown (1984; Brown & Palincsar, 1982) also reported impressive reading comprehension gains for elementary school subjects using a reciprocal teaching strategy which included self-questioning as a major component. By way of explanation of these effects, Brown and

Palincsar (1988) pointed out that in cooperative questioning contexts such as those used in the studies cited above, the group provides mutual support, shared expertise, and models for expert questioning and responding. These few studies suggest that a cooperative learning context may enhance comprehension effects of the self-questioning strategy.

Agency

Another question that arises is whether use of a self-questioning strategy also makes learners feel more in control of their own learning. Wang (1983; Wang & Lindvall, 1984), for example, has argued that use of metacognitive strategies enhances learners' sense of personal agency, that is, their perception of personal control. This effect is presumably due to the self-regulation or self-control aspect of metacognitions. Thus, since self-questioning is a metacognitive strategy, learners who use self-questioning might be expected to feel more in control of their learning than would those who use other comprehension strategies. Furthermore, McCombs (1988) has noted that research and theory support the view that increasing learners' sense of personal agency can also increase achievement (e.g., Bandura, 1982; King, 1983; Thomas, 1980; Wang, 1983). Personal agency refers to one's sense of self-determination or personal causation. Those who have a strong sense of agency perceive themselves to be personally empowered—drawing their power from within. High agency individuals feel in control of their lives; they determine their own goals and rewards, they create their own experience, and they have a strong sense of responsibility for that experience (King, 1983; Thomas, 1980).

The purpose of the study reported here was to investigate the effects of self-questioning and review strategies (used in both cooperative and individualistic learning environments) on college students' lecture comprehension and on their feelings of control over their own learning. Consistent with findings from studies using self-questioning strategies with *written* materials, participants who practiced with the self-questioning strategies were expected to show better comprehension than those who practiced with the review strategies (despite the possibility of any additional cognitive burdens such a strategy might place on the learner). Students who used the self-questioning strategies were also expected to show more improvement in lecture comprehension over time than those who used review. In addition, participants in a cooperatively structured self-questioning context were expected to achieve higher scores on comprehension measures than self-questioners who worked independently, consistent with previous research (Brown & Palincsar, 1982; Fantuzzo *et al.*, 1987; Palincsar & Brown, 1984). Quality of self-generated student questions was expected to increase or remain the same (but not decrease) over the duration of the study. Such a finding would demonstrate the stability

of participants' use of the self-questioning strategy. Finally, since self-questioning is a metacognitive strategy, students using this strategy were expected to feel more in control of their own learning and show a post-treatment increase in their sense of agency.

METHOD

Sample

Thirty-two college students enrolled in two sections of an elementary education course participated in the study. The mean age and grade point average (GPA) of the total group of students were 25.5 and 3.17, respectively. All participants were female with the exception of two males in one of the classes. Participants were graduate and upper-level undergraduates. The approximate ratio of graduate students to undergraduates was three to one in both classes.

Procedure

One of the two classes was randomly assigned to the self-questioning treatment and the other to the review treatment. Students within each of these two groups were then randomly assigned to either a cooperative or an individualistic learning context, and within the cooperative conditions participants were randomly grouped into triads. This procedure resulted in four treatment groups: cooperative self-questioning (hereafter called self- and peer-questioning—S/P-Q, $n = 9$), independent self-questioning, (S-Q, $n = 9$), cooperative review (CR, $n = 9$), and independent review (IR, $n = 5$).

To increase external validity, the study was embedded within the structure of an existing college course. The study was designed in such a way that strategy training, practice with learning strategies, and testing were incorporated into the course as a means of learning the regular course content. In this way all elements of the study became naturally occurring classroom phenomena which extended over the 3-month semester.

Participants were given pre- and post-treatment tests on agency and lecture comprehension. They also participated in four strategy practice sessions in which they listened to a lecture, studied the lecture content using their assigned strategy, and then took a comprehension test on the lecture material. Thus the research design consisted of four strategy groups with repeated measures on agency (pre- and post-treatment) and comprehension (pretreatment, practice, and post-treatment).

Participants were told that they would be divided into groups and would be using different learning strategies. Students in the two classes did not know each other and were asked not to discuss the strategies they were using with students outside the class.

Both classes were taught by the same professor, and throughout the duration of the study, the regular course content was presented in the same manner to the two classes. To maintain consistency between the two classes with regard to content covered, use of examples, and time allocated to topics, written lecture notes were rigidly followed for all presentations and students were not allowed to ask questions or interrupt during the experimental lectures. These procedures helped to ensure that the in-class experience of the four treatment groups was essentially the same and differed only with regard to learning strategy used by students.

Lecture presentations. Six of the course lectures were selected for use in the study. The first lecture was followed immediately by a comprehension pretest to gather baseline lecture comprehension data, the next four lectures were used for the practice sessions, and the sixth for the post-test session. The criteria for selection of these presentations were that they represented pure lecture (i.e., no activity groups, microteaching, or videos), and their content was unrelated to material covered in other courses (to avoid effects of prior knowledge

at testing). The six lectures were presented in logical order, following the usual topic sequence for this course. The lectures covered the following regular course topics: discovery learning, instructional planning, critical thinking, the concept-attainment teaching strategy, evaluation of student learning, motivation, and classroom discipline theories and practices.

Instruction in question generation and metacognition. All participants in the study were provided with direct instruction in both questioning skills and metacognitive principles as part of the regular course content. (These topics are appropriate for education majors and are generally covered in beginning education courses.) All participants learned to write a variety of question types including factual and higher-order questions of the sort generally used for instruction and testing in elementary classrooms. Factual questions were identified as ones which require the learner to simply recall facts and ideas explicitly stated, whereas higher-order questions were defined as requiring the learner to recall the facts and ideas and also to engage in application, analysis, interpretation, or evaluation of those ideas. Hamaker (1986) has argued that factual questions elicit rote learning while higher-order questions tend to elicit meaningful learning. All participants in the study were also provided with a brief explanation of metacognition including a discussion of the beneficial effects of monitoring one's comprehension during learning. Participants in all strategy groups were taught skills of generating questions as well as general concepts of metacognition to control for effects due to these skills and concepts.

Self-questioning strategy training. Students in the two self-questioning treatment groups (S/P-Q & S-Q) received additional training in that they were taught to *apply* their questioning skills and their knowledge of metacognition in a learning context by using *self*-questioning as a learning strategy. This is an example of what Brown and Palincsar (1982) have termed "informed strategy training," that is, informing the participants of the reason for using the strategy and helping them to see the relationship between strategy use and improved learning (Pressley, Levin, & Ghatala, 1988).

Question stems were used as a way of effectively training these students in self-questioning. Bean (1985) has pointed out that when self-questioning strategies are too complex, learners tend to modify them, adapt them, or use other learning strategies; therefore, if students are expected to use a self-questioning strategy and to incorporate it into their repertoire of strategies, the strategy needs to be uncomplicated and easy to internalize. The use of question stems appeared to fit these requirements. Students in the self-questioning condition were provided with a handout listing a set of generic "question starters" which they could use to guide them in asking higher-order comprehension questions specific to any presented content. The self-questioning training consisted of three 15-min sessions during which students practiced using these stems to write specific higher-order questions. The students in the self-questioning condition also practiced generating specific comprehension-monitoring (metacognitive) questions. During training participants shared with the class examples of the questions that they wrote. This sharing plus feedback provided ample modeling of well-constructed higher-order questions. (In pilot work with similar subjects this amount and type of training was sufficient for participants to reach proficiency in self-questioning.) Inspection of Table 1, which contains examples of question stems with sample student endings, shows that the question stems were designed to facilitate students' understanding of the lecture content, integration with prior knowledge, and comprehension-monitoring; that is, the question stems address both cognitive and metacognitive processes.

Strategy practice. Following each of the lectures in the practice and post-test sessions students engaged in a 10- to 12-min "study session" during which they reviewed the material presented. In pilot work, 12 min was found to be adequate for this group task. After each study session a comprehension test was administered. This sequence of lecture, study, testing was followed for each of the four practice sessions and the post-test session. During studying, each of the four treatment groups utilized only its assigned learning strategy: that

TABLE 1
QUESTION STEMS AND SAMPLE QUESTIONS GENERATED BY PARTICIPANTS

How is . . . related to . . .?
What is a new example of . . .?
In your opinion what is the best . . . and why?
What are some possible solutions for the problem of . . .?
Explain why . . .
What conclusions can you draw about . . .?
What is the difference between . . . and . . .?
How are . . . and . . . similar?
How would you use . . . to . . .?
What do you think would happen if . . .?
What are the strengths and weaknesses of . . .?
Why is it important . . .?
Do you agree or disagree with this statement: . . .? Support your answer.
What do I (you) still not understand about . . .?
How is (the concept attainment teaching strategy) related to (discovery learning)?
What is a new example of (Kounin's idea of "withitness")?
In your opinion what is the best (way to evaluate kids' essays, checklist or wholistic) and why?
What are some possible solutions for the problem of (working with kids who are used to getting M and M's all the time and you don't believe in using behavior modification)?
Explain why (you should use multiple measures when evaluating students' learning and give an example).
What conclusions can you draw about (the fact that there are so many different theories of motivation and discipline in the classroom)?
What is the difference between (discipline) and (classroom management)?
How are (critical thinking) and (creative thinking) similar?
How would you use (the critical thinking taxonomy we learned about) to (teach a lesson in art)?
What do you think would happen if (you let your fifth graders choose what they were going to learn about today)?
What are the strengths and weaknesses of (assertive discipline)?
Why is it important (to make kids support their inferences and conclusions)?
Do you agree or disagree with this statement: ("If you motivate your students you won't have discipline problems.")? Support your answer.
What do I (you) still not understand about (intrinsic motivation)?

is, self- and peer-questioning in small cooperative groups, self-questioning working alone, review in small cooperative peer groups, or independent review of the material.

Participants in both of the self-questioning conditions were instructed to spend the first couple of minutes of study time asking themselves two or three higher-order questions on the content presented. The S/P-Q treatment group was told to then use these questions in their cooperative learning triads to engage in reciprocal peer questioning and responding to those questions. The independent self-questioners, after generating their list of questions, proceeded to answer them, working independently all the time. Students in the review triads were instructed to engage in discussion of the material presented, whereas those in the independent review treatment were told to work alone to review the material by mentally

recalling the material, or in whatever manner they wished. During the study time all participants were allowed to consult any notes taken during the lecture.

Measures

Comprehension. A pretest was administered as a measure of students' ability to comprehend and remember lecture material. After each of the four practice sessions and the final session, a similar written test was administered to evaluate participants' comprehension of the content of that specific lecture. The final test served as the treatment post-test. These five comprehension tests were independent of each other in terms of content; that is, later tests did not build upon material from earlier lectures. All of the tests were administered immediately after the lecture and study session. All tests contained both multiple choice and open-ended questions. The difficulty level of test items was set at a high level to avoid the possibility of a ceiling effect. Test questions elicited higher-order thinking, requiring the learner to integrate several pieces of information, elaborate on ideas, analyze information, or apply it in a new context. Examples of items from the final posttreatment test are as follows:

The school district superintendent wants an overall picture of the level of achievement for each class in the school system. The best technique/ instrument to use is:
(a) rating scale
(b) teacher-made test
(c) standardized test
(d) checklist

Describe and defend two different ways to evaluate learning for the following classroom scenario: A sixth grade teacher has just finished teaching her students to compute the volume of a cube and wants to know how well they have learned this skill.

Using the lecture as an indicator of content covered, two independent raters, blind to student identity and experimental condition, scored the open-ended questions on each of the six tests. Interrater reliabilities on the tests ranged from .91–.95. In all cases of discrepant scores, an average was taken. The internal consistency (coefficient α) of the six comprehension tests ranged from .72 to .79. The level of internal consistency is well within the acceptable reliability range for experimental research (Gronlund, 1977).

Agency. Agency was measured by an instrument developed by the investigator specifically to measure the degree to which education students believe that they are the cause of their own success as learners and future teachers. The agency scale consists of eight Likert-type items. A high score on this instrument indicates a strong sense of agency. Reliability coefficient α for this instrument is .72. The agency scale was administered both pre- and post-treatment. Sample items from this scale follow:

How confident do you feel about your ability to understand college lectures?

very confident	somewhat confident	not very confident	not at all confident

My success as a teacher will depend heavily on the school that I happen to be assigned to and on the kind of principal I get.

strongly agree	agree	disagree	strongly disagree

RESULTS

One-way analyses of variance on the pretest data revealed no significant differences among the four groups for either age, GPA, agency, or lecture comprehension, suggesting initial comparability of the groups with regard to these variables.

Comprehension

Effects of practice with the four different strategies over time is graphically displayed in Fig. 1. Inspection of Fig. 1 reveals that, although comprehension means for the four groups were not significantly different at pretest, differences in performance among the groups became apparent in the first practice session and continued through to the post-treatment test. Specifically, the learning curves of the S/P-Q and S-Q groups appear to closely parallel each other over the six testing times; in addition, performance of these two self-questioning groups is consistently above that of both the cooperative review and independent review groups.

To statistically assess effects of practice with the four strategies over time, a 4(strategy) × 6(time) repeated measures ANOVA was conducted on participants' scores on the six comprehension tests. This analysis revealed significant effects for both strategy, $F(3,28) = 9.85$, $p < .001$, and time, $F(5,140) = 15.22$, $p < .001$, and a marginally significant interaction, $F(15,140) = 1.64$, $p < .07$.

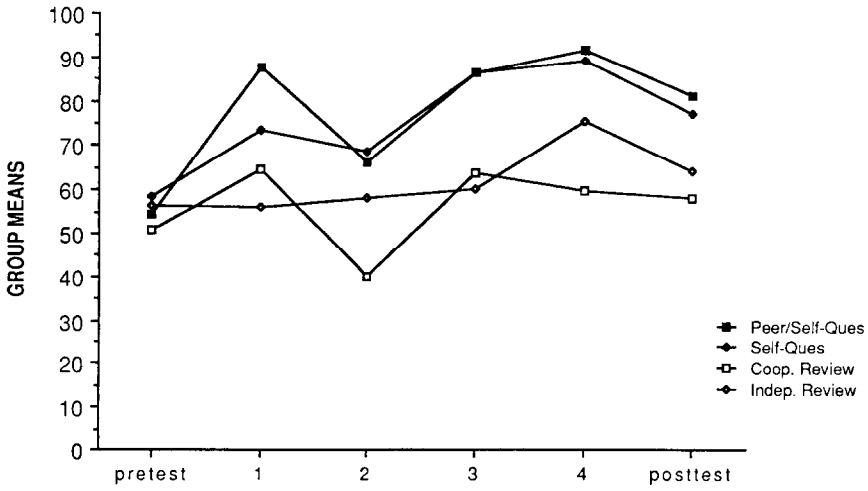


FIG. 1. Mean percentages on comprehension tests.

TABLE 2
GROUP MEANS FOR COMPREHENSION AT PRETEST, PRACTICE, AND POST-TREATMENT

	Strategy				
	Self- and peer-questioning	Self-questioning	Cooperative review	Independent review	Significant post-hoc analyses
Pretest	54.4	58.6	50.8	56.6	
Practice 1	87.6	73.6	64.4	56.0	P/S-Q > CR; IR
Practice 2	66.2	68.8	40.0	58.0	P/S-Q; S-Q > CR
Practice 3	86.2	86.1	63.9	60.0	P/S-Q; S-Q > CR; IR
Practice 4	91.6	89.0	60.0	75.6	P/S-Q; S-Q > IR > CR
Post-treatment	81.0	77.1	58.2	64.0	P/S-Q; S-Q > CR; IR

Inspection of Fig. 1 suggests that the significant effect for strategy was due to the performance of the two self-questioning groups. To provide a direct test of the hypotheses that the two self-questioning strategy groups would show more improvement in comprehension over time than the review strategy groups, and that these self-questioning groups would demonstrate comprehension consistently superior to that of the review groups, planned post-hoc comparisons were made within each of the four strategy groups (pre–post) and among the four groups at individual practice and post-test times. Comprehension means[1] for the four groups on the pretest, the four practice session tests, and the post-treatment test are presented in Table 2.

Correlated t tests (two-tailed) on pairs of pre- and post-test means for each of the four strategy groups revealed that the S/P-Q strategy group significantly improved comprehension over time, $t(8) = 3.89$, $p < .01$, as did the S-Q strategy group, $t(8) = 3.79$, $p < .01$, whereas improvement was marginal for the cooperative review group, $t(8) = 2.19$, $p < .07$, and nonsignificant for the independent review group, $p > .10$. These findings indicate that only the two self-questioning strategies significantly improved lecture comprehension over time.

To examine differences among the four strategy groups, post-hoc comparisons among group means at pretest, practice, and post-test times were conducted using t tests (two-tailed) and significant differences are displayed in Table 2. At pretest there were no significant differences; however, beginning with practice session 1 differences emerged with the S/P-Q group scoring significantly better than both the CR group, $t(16) = 2.83$, $p < .03$, and the IR group, $t(12) = 2.93$, $p < .01$. By practice session

[1]All comprehension test scores were converted to proportions prior to analysis.

3, four of the six comparisons were significant: again the S/P-Q group scored significantly better than both the CR group, $t(16) = 2.79$, $p < .01$, and the IR group, $t(12) = 3.01$, $p < .01$; the S-Q group also did better than both the CR group, $t(16) = 2.97$, $p < .01$, and the IR group, $t(12) = 3.40$, $p < .01$. The same four comparisons remained significant through practice session four and post-testing with all ps $< .01$. These results suggest that the self/peer-questioners as well as the self-questioners showed comprehension consistently superior to that of participants using either of the review strategies.

No significant differences were found between the S/P-Q and S-Q groups on lecture comprehension at any of the testing times (all ts$(8) < 1$).

The independent review group's performance with their strategy remained fairly stable over time from pre- to post-test except for a sudden rise on practice test 4. Indeed their performance was substantially better than that of the cooperative review strategy group in practice sessions 2, $t(12) = -2.58$, $p < .02$, and 4, $t(12) = -2.16$, $p < .06$.

Agency

A 4 × 2 repeated measures analysis of variance was used to analyze scores on the agency scale where strategy was the between-groups factor and time was the repeated measure. No significant differences in agency were found for either strategy, $F(3,28) < 1$, or time, $F(1,28) = 1.24$. The interaction was also nonsignificant, $F(3,28) = 1.49$. This finding indicates that students' sense of agency was not changed as a result of using any of the four strategies.

Questions

To determine whether the *quality* of questions generated changed over time for cooperative vs independent self-questioners, students' questions from each of the study sessions were rated by two independent raters for overall quality. A question's quality was judged in terms of the extent to which it required the learner to apply, interpret, analyze, or evaluate lecture content. Average interrater reliability for the five sets of questions was .92. A repeated measures analysis of variance on the ratings of questions generated during practice sessions revealed no significant differences between the self-questioning strategy groups ($F < 1$) and no change over time for either group ($F < 1$).

DISCUSSION

Findings of this study indicate that learners can benefit from using a

self-questioning strategy to improve their comprehension of lectures. Participants in the self-questioning conditions comprehended the lecture material significantly better than did those in the review groups. Thus, results of this study are consistent with research on the use of self-questioning in *reading* comprehension (e.g., Bean, 1985; Wong, 1985). However, this study contributes to the literature on self-questioning by extending these findings to a broader context of classroom learning which includes lectures.

Metacognition

As has been the case with reading comprehension, the success of the self-questioning strategy with *lecture* comprehension can probably be attributed to the metacognitive aspects of self-questioning. In this study participants in all four strategy groups were trained in skills of writing higher-order questions, and all students learned basic principles of metacognition including the benefits of comprehension-monitoring during learning. However, the strategy treatments differed in that the self-questioners received direct instruction in how to *apply* this information by using *self*-questioning and they were also provided with the metacognitive rationale for use of this strategy; that is, they engaged in *informed* strategy training (Brown & Palincsar, 1982). Informed training has been found to be superior to blind training in a number of comprehension studies (e.g., Brown & Palincsar, 1982; Paris, Cross, & Lipson, 1984; Pressley, Borkowski, & O'Sullivan, 1984); however, in this case it is difficult to say how much of the observed effect was due to use of the strategy per se and how much to being informed of the strategy's rationale. This issue could be explored by comparing the effects of self-questioning training with *informed* self-questioning training.

It is possible that the self-questioning training improved comprehension performance because the training actually influenced how students encoded the material *during* the lectures. Such self-questioning training could have affected how students approached the lecture. For example, knowing that they would have to generate questions following the lecture may have given rise to students' question generation *during* the lecture, and this in turn may have influenced their encoding process. Results of a related study in which high school students were specifically instructed to generate their questions *during* the lectures (King, in press) lend support to this notion.

Cooperative Learning

It is not clear exactly why the self-questioning strategy was equally effective in individualistic and cooperative contexts. One possibility is

that the mutual peer support function of the group did not play as prominent a role in learning as both Brown and Palincsar (1988) and Fantuzzo *et al.* (1987) have suggested it does. On the other hand, it is also possible that the use of the question stems provided the independent learners with the modeling benefits that Brown and Palincsar (1988) found to be so important in the peer questioning context, and this benefit may have offset the peer support advantages of the cooperative condition. Further research on effects of this self-questioning strategy should attempt to isolate the modeling and peer support variables.

It is difficult to say why the performance curve for the cooperative review group appears more unstable than that of the independent review group. Through years of schooling these college age students most likely had developed their own most effective strategies which worked well for them. This may account for why the independent review students performed fairly consistently across all tests. Also these mature students were probably more adept at studying alone than in groups and, for the cooperative review students, an attempt to learn a new strategy in cooperative groups probably interfered with their learning proficiency. Perhaps younger less-experienced students would not have such difficulty.

Agency

Results of this study did not support the expectation that college students who use the self-questioning strategy will feel more in control of their own learning. Scores on the agency scale did not increase significantly in any of the strategy conditions. Possibly the social aspects of the cooperative groups interfered with the *self* emphasis of the metacognitive strategy. On the other hand, the agency scale measured a fairly general sense of agency which may not be easily modified in such a short time.

Strategy Stability

A question that may arise is whether the quality of self-questioning deteriorated during the course of the study after the novelty wore off. This question was of serious concern because, although students in the self-questioning conditions were allowed to use the question stems handout as a prompt during question generation, most indicated that they no longer needed the handout by the second session. Ratings of the type and quality of questions generated revealed that variety and quality of questions had not diminished over time for either group. This finding indicates that the question stems were quickly internalized by the learners and remained stable over the course of the study and further suggests that this strategy may be relatively resistant to the kinds of learner modifications that Bean (1985) warned about.

Although the self/peer-questioning and self-questioning strategies appear to improve college students' comprehension of lectures on course material related to the teaching profession, these findings are not necessarily generalizable to other learning contexts. Further research needs to be conducted using these strategies with other age groups and subject matter to determine their general effectiveness. Results from a study comparing the effectiveness of these same strategies on high school students' comprehension of history lectures (King, in press) lend support to the comprehension findings from the college student sample; however, these strategies may not be effective in other learning contexts. For example, students in the elementary grades may not be able to readily formulate higher-order questions. Moreover, higher-order questioning may be appropriate for some subject areas but not for others. Subject variables such as ability level and learning style may also influence the effectiveness of self-questioning as a lecture comprehension strategy. Additional research in a variety of learning contexts with elementary, high school, and college students is necessary to explore these issues.

Finally, it should be noted that the use of self-questioning in this study suggests that such a strategy can be successfully incorporated into ongoing classroom instruction or similar real world learning contexts. Specifically, in this study use of the self-questioning and self/peer-questioning strategies became a regular part of the course and, according to informal student feedback, were not perceived as obtrusive or unnatural. Students' comments during the debriefing session at the end of the study indicated that these college students found the self-questioning strategy so beneficial and easy to apply that they intended to continue using it to help them learn from lectures.

REFERENCES

ANDRE, M. D. A., & ANDERSON, T. H. (1978). The development and evaluation of a self-questioning technique. *Reading Research Quarterly*, **14**, 605–623.

BANDURA, A. (1982). Self-efficacy mechanism in human agency. *American Psychologist*, 33, 344–358.

BEAN, T. W. (1985). Classroom questioning strategies: Directions for applied research. In A. Graesser & J. B. Black (Eds.), *The psychology of questions*. (pp. 335–358). Hillsdale, NJ: Erlbaum.

BROWN, A. L. (1980). Metacognitive development and reading. In R. L. Spiro, B. B. Bruce, & W. F. Brewer (Eds.), *Theoretical issues in reading comprehension* (pp. 453–481). Hillsdale, NJ: Erlbaum.

BROWN, A. L., & PALINCSAR, A. S. (1982). Inducing strategic learning from texts by means of informed, self-control training. *Topics in Learning and Learning Disabilities*, **2**(1): 1–17.

BROWN, A. L., & PALINCSAR, A. S. (1988). Guided, cooperative learning and individual knowledge acquisition. In L. Resnick (Ed.), *Cognition and instruction: Issues and agendas*. Hillsdale, NJ: Erlbaum.

DANKS, J. H., & END, L. J. (1987). Processing strategies for reading and listening. In R. Horowitz & S. J. Samuels (Eds.), *Comprehending oral and written language*. New York: Academic Press.

DAVEY, B., & MCBRIDE, S. (1986). Effects of question-generation training on reading comprehension. *Journal of Educational Psychology*, **78**, 256–262.

FANTUZZO, J. M., DIMEFF, L. A., & FOX, S. L. (1987). *Reciprocal peer tutoring: A multimodal assessment of effectiveness*. Paper presented at the annual meeting of the American Psychological Association, New York.

FLAVELL, J. H. (1981). Cognitive monitoring. In W. P. Dickson (Ed.), *Children's oral communication skills*. New York: Academic Press.

FRASE, L. T., & SCHWARTZ, B. J. (1975). Effect of question production and answering on prose recall. *Journal of Educational Psychology*, **67**, 628–635.

GRONLUND, N. E. (1977). *Constructing achievement tests*. Englewood Cliffs, NJ: Prentice-Hall.

HALLER, E. P., CHILD, D. A., & WALBERG, H. J. (1988). Can comprehension be taught? *Educational Researcher*, **17**(9), 5–8.

HAMAKER, C. (1986). The effects of adjunct questions on prose learning. *Review of Educational Research*, **56**, 212–242.

HERTZ-LAZAROWITZ, R. (1985). Internal dynamics of cooperative learning. In R. Slavin, S. Sharan, S. Kagan, R. Hertz-Lazarowitz, C. Webb, & R. Schmuck, (Eds.), *Learning to cooperate, cooperating to learn* (pp. 97–102). New York: Plenum.

JOHNSON, D. W., MARUYAMA, G., JOHNSON, R., NELSON, D., & SKON, L. (1981). Effects of cooperative, competitive, and individualistic goal structures on achievement: A meta-analysis. *Psychological Bulletin*, **89**, 47–62.

KING, A. (1983). Agency, achievement, and self-concept of young adolescents. *Studies in Art Education*, **28**, 187–194.

KING, A. (In press). Improving lecture comprehension: Effects of a metacognitive strategy. *Applied Cognitive Psychology*.

MARKMAN, E. M., & GORIN, L. (1981). Children's ability to adjust their standards for evaluating comprehension. *Journal of Educational Psychology*, **73**, 320–325.

MCCOMBS, B. L. (1988). Motivational skills training: Combining metacognitive, cognitive and affective learning strategies. In C. E. Weinstein, E. T. Goetz, & P. A. Alexander (Eds.), *Learning and study strategies*. New York: Academic Press.

NOLTE, R., & SINGER, H. (1985). Active comprehension: Teaching a process of reading comprehension and its effects on reading achievement. *The Reading Teacher*, October, 24–31.

PARIS, S., CROSS, D., & LIPSON, M. (1984). Informed strategies for learning: A program to improve children's reading awareness and comprehension. *Journal of Educational Psychology*, **76**, 1239–1252.

PALINCSAR, A. S., & BROWN, A. L. (1984). Reciprocal teaching of comprehension-fostering and monitoring activities. *Cognition and Instruction*, **1**(2), 117–175.

PRESSLEY, M., BORKOWSKI, J. G., & O'SULLIVAN, J. T. (1984). Memory strategy instruction is made of this: Metamemory and durable strategy use. *Educational Psychologist*, **19**, 94–107.

PRESSLEY, M., LEVIN, J. R., & GHATALA, E. S. (1988). Strategy-comparison opportunities promote long-term strategy use. *Contemporary Educational Psychology*, **13**, 157–168.

SHARAN, S. (1980). Cooperative learning in small groups: Recent methods and effects on achievement, attitudes, and ethnic relations. *Review of Educational Research*, **50**, 241–271.

SINGER, H., & DONLAN, D. (1982). Active comprehension: Problem-solving schema with

question generation for comprehension of complex short stories. *Reading Research Quarterly*, **17**, 166–186.

THOMAS, J. W. (1980). Agency and achievement: Self-management and self-regard. *Review of Educational Research*, **50**, 213–240.

TOWNSEND, D. J., CARRITHERS, C., & BEVER, T. G. (1987). Listening and reading processes in college- and middle school-age readers. In R. Horowitz & S. J. Samuels (Eds.), *Comprehending oral and written language*. New York: Academic Press.

WANG, M. C. (1983). Development and consequences of students' sense of personal control. In J. M. Levine & M. C. Wang (Eds.), *Teacher and student perceptions: Implications for learning*. Hillsdale, NJ: Erlbaum.

WANG, M. C., & LINDVALL, C. M. (1984). Individual differences and school learning environments. In *Review of research in education* (Vol. 2). Washington, DC: American Educational Research Association.

WONG, B. Y. L. (1985). Self-questioning instructional research: A review. *Review of Educational Research*, **55**, 227–268.

WONG, B. Y. L. (1986). The efficacy of a self-questioning summarization strategy for use by underachievers and learning-disabled adolescents in social studies. *Learning Disabilities Focus*, **2**(1), 20–35.